

Molecular Understanding of Bilayer Structural Transition of CO₂ at the Ionic Liquid–Carbon Nanotube Electrode Interface

Hongwei Chen, Yiming Han, Xiaoxiao Hu, Xiang Wei,* Yangfan Song, and Zhuo Liu



Cite This: *J. Phys. Chem. C* 2024, 128, 8426–8439



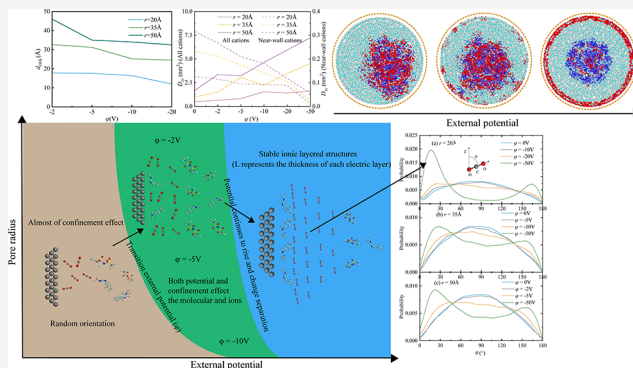
Read Online

ACCESS |

Metrics & More

Article Recommendations

ABSTRACT: In this paper, the atomistic-level descriptions of the electric double layer (EDL) formation of a CO₂/ionic liquid (IL) on the carbon nanotube electrode are revealed by molecular dynamics simulations. The interfacial microenvironment and the orientation distribution depending on the pore radius and external potential are investigated. Throughout the evolution process, three different stages can be recognized by the variations of the coordination number and density distribution of cations. For small pore sizes, the interfacial structure is initially dominated by the nonelectrical force caused by the confinement effect; therefore, a similar density distribution of cation–anion ion pairs is observed in the vicinity of the electrode. With the increase of external potential, the cation [Bmim⁺] aggregates near the electrode, and the ion pairs are gradually separated by the CO₂ molecule. Meanwhile, the coordination microenvironment of [Bmim⁺] is gradually occupied by the combination of [Bmim⁺] and CO₂, and a steady EDL structure is formed. Once the external potential exceeds the critical value, the spontaneous charge separation phenomena can be determined by the interfacial density. With the decrease of pore radius, the critical potential for the transition becomes earlier. The EDL formation under potential polarization is associated with the simultaneous rearrangement of the orientation and mobility of [Bmim⁺] and CO₂. The mobility and the orientation distribution determined from the MD simulations coincide well with the transition process. Once the transition is initiated, the orientation of CO₂ changes from disordered to parallel to the electrode, while the imidazole ring of [Bmim⁺] is gradually oriented preferentially perpendicular to the electrode surface. On the other hand, the self-diffusion coefficient begins to decrease when structural transition is initiated. Once the transition process is completed, the value increases again, which reaffirms the microstructural evolution of EDL.



1. INTRODUCTION

Capturing carbon dioxide by electrochemical cells is considered to be a promising technology for constructing sustainable energy systems in the future. Because of the excellent electrical conductivity and terrific reactive area, carbon nanotubes are broadly used as a supporting base in the field of energy storage and electrode materials in cells and rechargeable batteries.¹ Electric double-layer capacitors (EDLC)² are high-density energy storage devices, which are widely used in energy storage and reduction reaction of CO₂ (CO₂RR).³ The CO₂RR performance relies primarily on both the electrocatalysts and the reaction environment surrounding the solid–liquid–gas interface.⁴ During the processes of CO₂RR, the electrolyte attaches to the electrode surface, forming an electrical double-layer structure that changes interfacial structure,⁵ which is related to the subsequent reaction process. Carbon nanotubes are used as carriers to increase the active sites of some organic catalysts,⁶ and ionic liquid (IL) are solvent-free electrolytes and have been regarded as an excellent choice for use in CO₂RR due to their wide

electrochemical window and stable physicochemical properties. Optimizing interfaces can be achieved by altering the interplay among factors, such as surface structure, interfacial electric field, and ion distribution, all situated at the molecular level. These factors collectively influence the thermodynamic and kinetic properties of electrocatalytic reactions. A comprehensive illustration of the CO₂–ionic liquid (CO₂–IL) interfaces is needed to find higher energy density and efficiency capture devices.

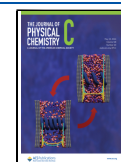
The interfaces have been classified as neutral interfaces and electrified ones.⁷ The former is important to understand the solvation and evolution structures of the CO₂–IL bilayer in the

Received: January 29, 2024

Revised: April 13, 2024

Accepted: April 30, 2024

Published: May 8, 2024



confined region, and the latter is quite crucial for electrochemical processes. Therefore, the evolution process of the electric double layer at the electrode surface is of particular importance for the development and optimization of energy utilization technology.^{8,9} The electric double-layer region is where the electron conductor contacts the electrolyte solution and where the electrochemical reaction occurs. Due to the synergistic effect of intermolecular interaction and induced field near the solid surface, different types of EDL are presented.¹⁰ In particular, IL mixtures have attracted more and more attention to regulate the microenvironment of the reaction interface and optimize the mass transfer process. Liang et al.¹¹ modeled the interfacial transition process by combining the particle transport equation and Poisson's formula to describe the complex morphological structure of the electrode–electrolyte interface. Ravikovitch et al.¹² used density functional theory (DFT) to obtain the pore structure and density of nanoporous catalysts through the calculation of adsorption isotherm, and the results indicated that many pores of different sizes and shapes existed on the surface of the catalysts, which indicated that the interfacial structure is affected by the pore size, causing a complicated evolution process.

Previously, the classical Gouy–Chapman–Stern (GCS) model^{13,14} has been put forward to predict the nature of the electrode/electrolyte interfaces. Oldham et al.¹⁵ found that the EDL in metal-ionic liquids system can be described by the GCS model. Gongadze et al.¹⁶ found that the subtle interactions between water sequences and ions and the external potential can be predicted by stern theory. Wang et al.¹⁷ investigated the transient electric double-layer dynamics at different concentration electrolyte solutions combined with the Poisson–Nernst–Planck (PNP) model. Various studies have shown that simulations could predict the shape of capacitance and current versus potential curves well at the atomic level. On the other hand, experiments with ionic liquids in confined geometries employing NMR and X-ray diffraction confirm vital findings. Islam et al.¹⁸ observed bell-shaped differential capacitance versus potential curves at the EDL structure at the electrode–electrolyte interface; results coincided with the Kornyshev electric double-layer structure. Tenney et al.¹⁹ simulated the results of density of nine ionic liquids at different pore sizes by comparing with the experimental results; they found that the pore size has an essential influence on the interface structure.

Nevertheless, most of the theoretical and experimental work cared about the evolution process of the ionic liquid interface, while the role of CO₂ in the CO₂–IL bilayer mixture is lacking. In addition, the interaction of confinement and electrostatic forces suggests that the evolution process differs from that of pure ionic liquid electrolytes.^{20,21} The mechanism of the action of CO₂ in bilayers may be more complex. Besides, the pore size can also influence the internal structure to a great extent. Jiang et al.²² pointed out that when the radius of a porous EDL capacitor is below 10 nm, the structure of the EDL is affected by a tightly packed structure of counterions and aligned dipole.

Different from the pore size effect, which is restricted by the microscale, MD can accurately describe the EDL structure and thermodynamic properties at an atomistic scale, which can be regarded as a potential supplement for the experiment. Many scholars have studied the synergistic process of confinement effect by MD simulations.²³ Yamada et al.²⁴ explored the impact of various pore sizes on the spatial distribution of water

molecules by MD simulations. Results showed that the oriented relaxation becomes lumpy when the distance is sufficiently large and the distance dependence of the spectral diffusion coefficient is independent of pore size. Winarto et al.²⁵ simulated the structural distribution of water molecules within carbon nanotubes, and the results revealed that the water molecules appeared to have a solid-wise structure because of the domain-limiting effect. Noh et al.²⁶ studied the charging phenomenon of an electric double-layer capacitor with ([EMIM]⁺[SCN][−]) ionic liquid solvents by MD methods. The results showed the interfacial structural distribution and identified the charging mechanism of the EDLC. Yang et al.²⁷ investigated the confinement effect at the entrance of the graphene layer by MD simulation, revealing the role of edges in ion separation and adsorption in micro-channels.

The difficulty of finding the thermodynamic characteristics of electric double layers in confined systems lacks a description of the electrostatic potential. The fixed charge method (FCM) and the constant potential method (CPM) are currently used in a large number of studies.²⁸ Among them, the FCM assumes the electrode atoms contain the constant charge, but it leads to charge fluctuations on the electrode because it ignores the fluctuations of the interfacial potential distribution.²⁹ To consider the effect of density fluctuations, Reed et al.³⁰ proposed the constant potential method, which applies a predetermined external potential at each simulation step so that it remains constant at a given electrode. Merlet et al.³¹ used CPM to study the positive and negative differential capacitance of a coarse-grained model at different external potentials; results showed that the calculated differential capacitances under CPM matched the simulation results very well under the all-atom methods. In a subsequent study, Merlet et al.³² used CPM to investigate the modeling of the adsorption of ionic liquids inside a microporous carbon electrode, and the ion adsorption process in porous structures can be well described.

In summary, the structural properties of the electrode interface change with the confinement effect; simultaneously, the evolution phase diagram of the interfacial structure of a binary mixture of CO₂ and ionic liquid is unclear, so revealing the effect of CO₂ in a mixture system is imminent. Through the distribution of potential and density in carbon nanotube internals, a systemic map of the interfacial structure and transformation process of the EDL structure are presented.

2. METHODS

Figure 1a shows the domain region of the CO₂–IL mixture system. The length of the carbon nanotube in the *z*-direction is set to 10 nm, which leads to the neglected edge effect. The CNT diameters ranging from 2 to 50 nm are excellent support materials for inducing strain in planar molecules smaller than 2 nm,³³ so pore radii of 20, 35, and 50 Å are selected to investigate the structural and orientational characteristics of IL/CO₂ within the CNT electrode. Both the CO₂ molecule and [Bmim]⁺ [Tf₂N][−] ionic liquid are enclosed inside the carbon nanotube electrodes. The structures of CO₂ and the ions are presented in Figure 1b. The boundary conditions in the *x*, *y*, and *z* directions are set to be periodic.

The interactions between carbon atoms at the carbon nanotube wall are modeled using the LJ potential.³⁴ The values ϵ_{ij} and σ_{ij} of carbon nanotube are 0.068 kcal/mol and 3.407 Å, while the values ϵ_{ij} and σ_{ij} of C/O in CO₂ are 0.439/0.878

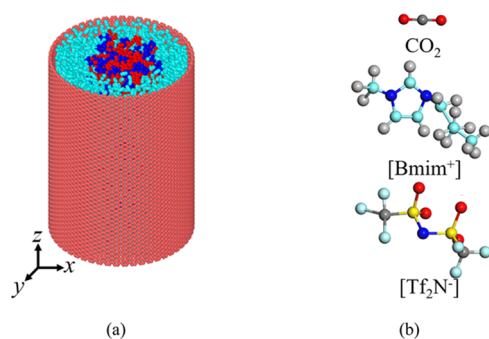


Figure 1. (a) simulation snapshots of the initial configurations and (b) schematic representation of [Bmim]⁺ and [Tf₂N][−] ions.

kcal/mol and 3.75/2.96 Å based on TraPPE force fields.³⁵ The atoms of IL are modeled using the all-atom OPLS model, which is widely used in the simulation of ionic liquid.^{36,37} The potential energy is expressed as

$$\begin{aligned}
 U_{\text{Cconf}} = & \sum_i \sum_{j>i} 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \sum_i \sum_{j>i} k \frac{q_i q_j}{r_{ij}} \\
 & + \sum_{\text{bond}} K_r (r - r_0)^2 + \sum_{\text{angle}} K_\theta (\theta - \theta_0)^2 \\
 & + \sum_{\text{bond}} \left\{ \frac{V_1}{2} [1 + \cos(\phi)] + \frac{V_2}{2} [1 + \cos(2\phi)] \right. \\
 & \left. + \frac{V_3}{2} [1 + \cos(3\phi)] \right\} + \sum_{\text{improper}} K_\chi (\chi - \chi_0)^2
 \end{aligned} \quad (1)$$

The first two terms represent the van der Waals force (VDW) and the Coulomb force, respectively. k is Coulomb's constant. After this, the bond stretching, angular bending, rotation around dihedral angles, and improper torsion angles are shown in sequence. The symbols of r , θ , ϕ , and χ represent the bond length, bond angle, intrinsic dihedral angle, and improper dihedral angle, respectively, ϵ_{ij} , σ_{ij} , K_r , r_0 , K_θ , θ_0 , V_1 , V_2 , V_3 , K_χ , and χ_0 are the force field coefficients, and q is the ionic charge. The interaction is determined by the Lorentz–Bertholet method.³⁸

Considering the influence of ion pair concentration, the typical density of 1.3 g/cm³ is chosen as mentioned in previous research.^{39,40} Initially, another simulation was conducted to determine the CO₂ capacity within carbon nanotubes, as illustrated in Figure 2a. CO₂ molecules surround the CNT/IL system, and the spontaneous CO₂ capacity is monitored, as shown in Figure 2b. The maximum number of CO₂ molecules entering the nanotubes is approximately 1500 at $r = 20$ Å. Subsequently, 1430 CO₂ molecules are selected at $r = 20$ Å, while the number of IL pairs is 65. To ignore the effect of concentration at different pore sizes, the relative proportions of CO₂/IL remain constant, which are 4400/200 and 9020/410 at 35 and 50 Å, respectively.

The motion of atoms is iterated by the velocity-Verlet method, with a time step of 1 fs. The system started at an external potential of 0 V under isothermal conditions at 300 K. 3 ns is used to obtain the equilibrium configuration, and the statistical data are collected in 1 ns by average.

The constant potential method is applied to produce a constant negative potential on the carbon nanotube. To

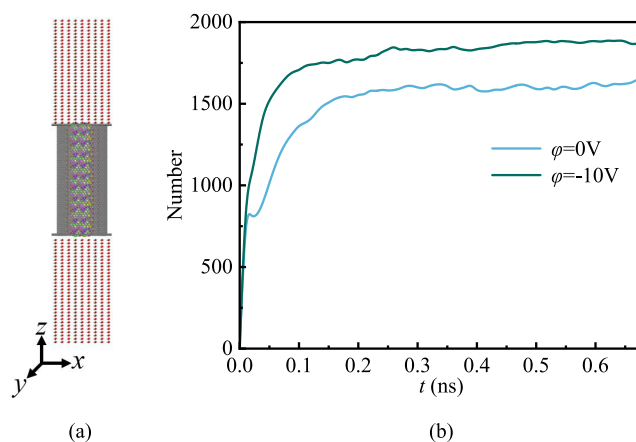


Figure 2. (a) Schematic diagram of the CO₂ filling of the CNT/IL system. (b) Maximum number of CO₂ inside carbon nanotubes with time.

explore the detailed structural patterns thoroughly, the typical external potential is set to 0V, −2, −5, −10, −20, and −50 V, respectively.^{41,42} It should be noted that the constant potential method allows for fluctuations in the atomic charge of carbon dioxide and ionic liquid inside the carbon nanotube, which is calculated by the Poisson equation.⁴³

$$\nabla^2 \phi = -\frac{\rho_e}{\epsilon_0} \quad (2)$$

where ϕ is the potential, ρ_e represents the charge density, and ϵ_0 is 8.854×10^{-12} F/m. For carbon nanotubes, we used the constant potential method. This means that the charge of each carbon atom on the wall is constantly changing. At each time step, we recalculated the charge of all atoms on the wall. Since the atoms inside the carbon nanotube are constantly moving in different time steps, this causes their positions to change. Figure 3a shows how the Coulomb force acts when a potential is applied. There are n atoms in the wall of the tube; each atom in the carbon nanotube is subjected to the force on the tube wall from the first atom to the n th atom, and all atoms on the wall of the carbon nanotube have the same charge in a single simulation by the CPM approach. The tube wall potential is examined, and a comparison between the external potential and the recalculated value is shown in Figure 3b. It can be found that the error between the calculated results and the external potential is less than 0.2% and can be verified constant potential method.

$$\phi_i = \frac{k \sum_{j \neq i} q_j}{r_{ij}} \quad (3)$$

In this model, the potential of each point charge is denoted by Φ , the Coulomb constant is denoted by k , the charge of each atom is denoted by q , and the distance from particle i to particle j is denoted by r_{ij} . To count the properties of the system, we sliced the system along the radius, with each layer having a thickness of 0.1 Å. The number of atoms in each layer is counted and then transformed to obtain the density expression as follows⁴⁴

$$\rho = \frac{N_{\text{bin}} M}{N_A V_{\text{bin}}} \quad (4)$$

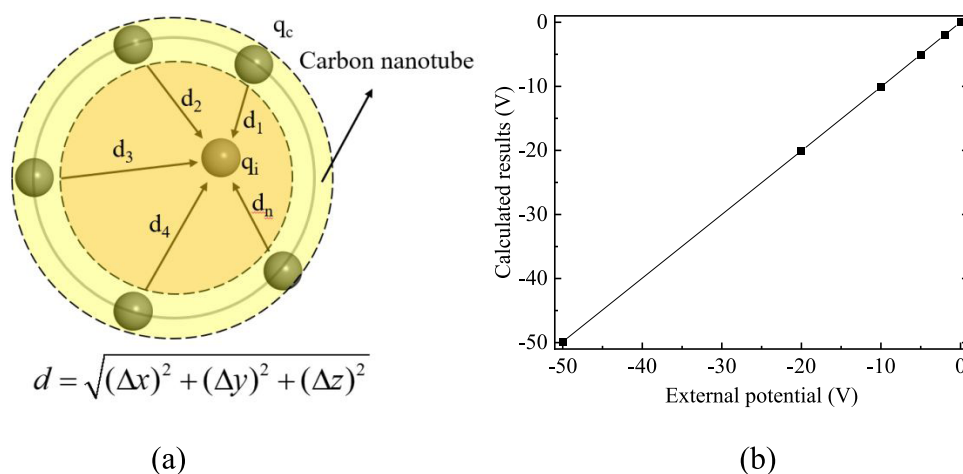


Figure 3. (a) Schematic diagram of the potential force. (b) Calculated results plotted against external potential.

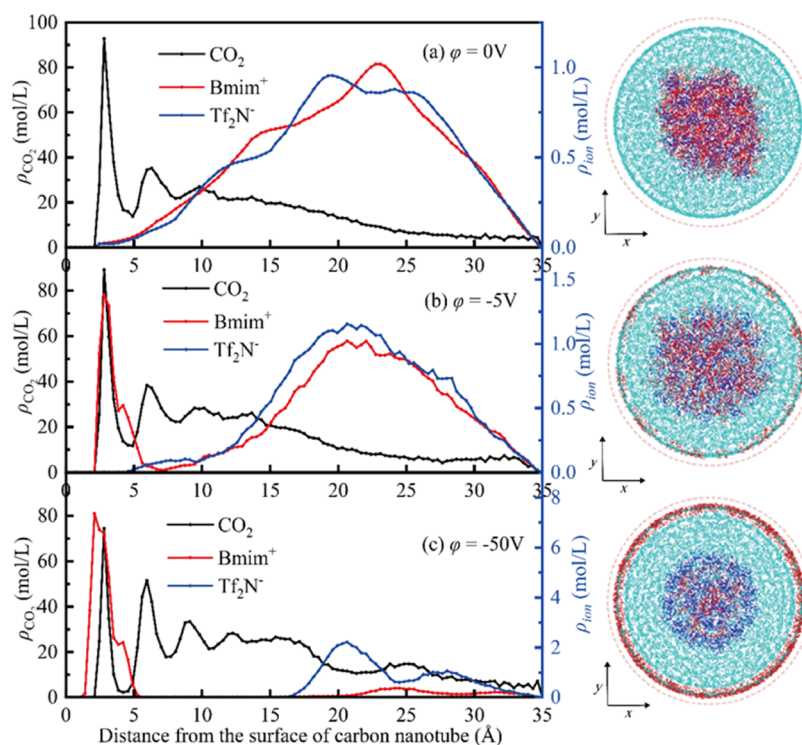


Figure 4. Density distribution and MD snapshots when $r = 35$ Å for the different external potential (a) $\phi = 0$ V, (b) $\phi = -5$ V, and (c) $\phi = -50$ V, respectively.

where M represents the relative mass, N_{bin} represents the number of particles in the sample region, V_{bin} is the volume, and N_A is Avogadro's constant. The radial distribution function and the coordination number are two important parameters for studying the microstructural ordering of substances, and their definitions are shown below.⁴⁵

$$g(r) = \frac{1}{\rho 4\pi r^2} \frac{\sum_{i=1}^r \sum_{j=1}^N \Delta N(r \rightarrow r + \delta r)}{N \times \tau} \quad (5)$$

$$N(r) = 2\pi\rho \int_0^r g(r)rdr \quad (6)$$

where N represents the number density, δr is the sample intervals, and ΔN is the amount of particles in the range r to δr . $N(r)$ represents the coordination number. The self-

diffusion coefficient D , determined by mean square displacement MSD, is used to characterize the change in motion of ions and molecules under the influence of external potential. Based on the Einstein relation, the MSD_{xy} and D_{xy} are calculated as follows⁴⁶

$$\text{MSD}_{xy}(t) = \langle |r_i(t) - r_i(0)|^2 \rangle \quad (7)$$

$$D_{xy} = \frac{1}{4} \frac{d(\text{MSD})}{dt} \quad (8)$$

where the MSD_{xy} was calculated for each i -particle at time t and D_{xy} is the self-diffusion coefficient in the two-dimensional (2D) plane.

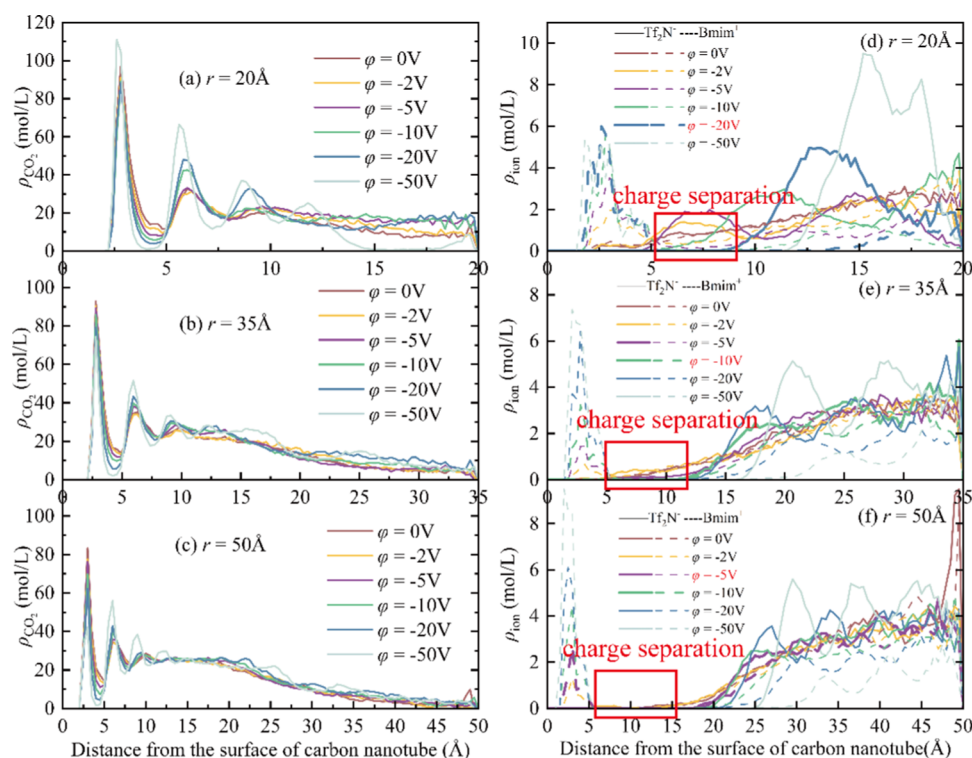


Figure 5. Density distribution of CO₂ at (a) $r = 20$ Å, (b) $r = 35$ Å, and (c) $r = 50$ Å and ions at (d) $r = 20$ Å, (e) $r = 35$ Å, and (f) $r = 50$ Å along the distance from the nanotube wall.

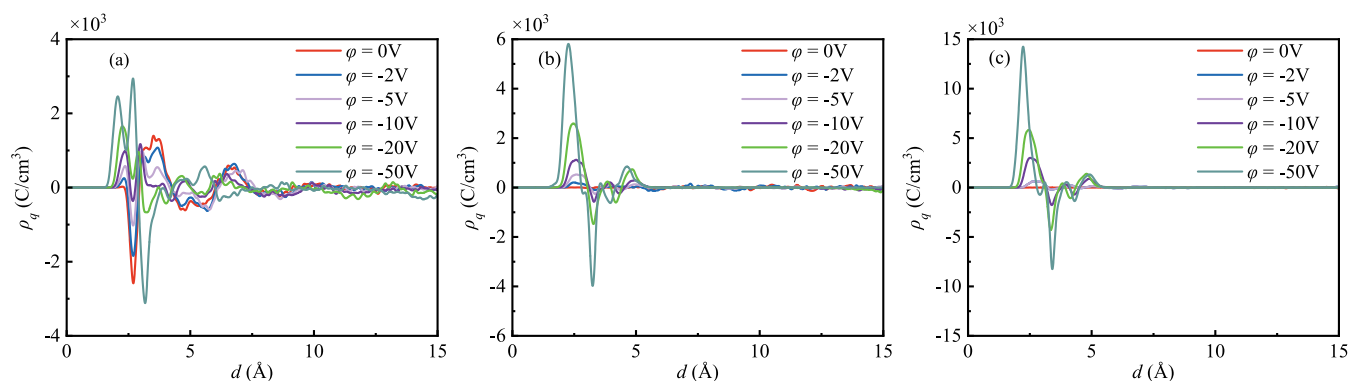


Figure 6. Charge density distribution along the distance from the pore wall when (a) $r = 20$ Å, (b) $r = 35$ Å, and (c) $r = 50$ Å.

3. RESULTS AND DISCUSSION

3.1. General Thermodynamics Properties. When the IL contacts the carbon nanotube, the thermodynamic behaviors of ions will be regulated by the external potential near the electrode, forming the electric double layer. Figure 4 depicts the typical density profiles varying with external potential at $r = 35$ Å. As presented in Figure 4a, for the initial state in the absence of an external potential, almost no density peak of ions appears near the electrode. When the distance away from the surface increases, a similar density distribution of [Tf₂N][−] anions and [Bmim]⁺ cations can be observed, and the density peak is found to be located at $d = 22.5$ Å. When the external potential increases to -5 V, the density profiles of the [Tf₂N][−] anion hardly change. But for the [Bmim]⁺ cation, it begins to accumulate near the electrode interface, and a density peak of 1.3 mol/L appears at $d = 2.5$ Å. With the external potential increasing further, the density profiles for the [Tf₂N][−] anion shift away from the electrode interface. Meanwhile, the density

peak for [Bmim]⁺ increases to 7.2 mol/L. Thus, the attraction force between ion pairs affected by the confinement effect vanishes, and the phenomena of charge separation happen, which have been confirmed in existing studies.⁴⁶ For the CO₂ molecules, the thickness of the adsorption layer near the electrode is enhanced dramatically with increasing external potential. As the external potential reaches -50 V, the thickness is over 20 Å, and the phenomenon of the layered arrangement can be observed, which is attributed to the intermolecular interactions. The MD snapshots are depicted to show the density profiles in the vicinity of the electrode surface intuitively. The outermost layer of light red color is the carbon nanotube, while the inside of the tube represents CO₂ molecules, [Bmim]⁺ cations, and [Tf₂N][−] anions modeled by blue, red, and navy colors, respectively. When the strength of the external potential is enlarged, some of the [Bmim]⁺ gradually accumulates to the electrode neighbor, while the CO₂ is squeezed out of the near-wall region. When the external

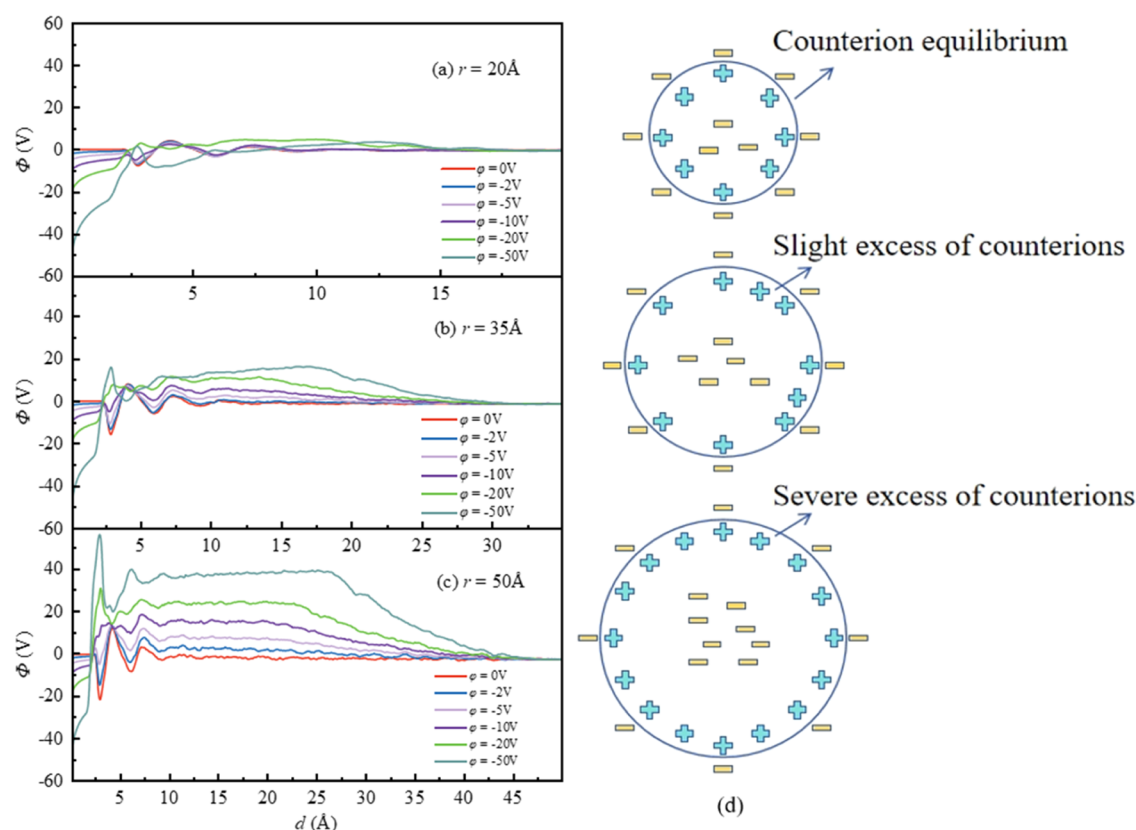


Figure 7. Distributions of potential in the tube along the radius direction of different pore radii (a) $r = 20$ Å, (b) $r = 35$ Å, and (c) $r = 50$ Å. (d) Schematic modeling of the distribution.

potential $\phi = -50$ V, the separation phenomena of heterologous ions can be observed.

The external potential of separation phenomena in the existence of CO_2 is shown in Figure 5. For the density distribution of CO_2 in Figure 5, the density of CO_2 changes little under different external potentials; once r varies from 20 to 50 Å, the maximum density peak changes from 110 to 81 mol/L due to the weakening of the confinement effect of the electrode. But for the ions, the trends are reversed. When the value of r equals 20 Å, the density peak of $[\text{Bmim}^+]$ is about 5.5 mol/L; with the increases of the external potential, the value enhances slightly. As $r = 35$ Å, the value reaches 6 mol/L, the overlap of nonelectrostatic force caused by the confinement effect is weakened, and the influence of external potential is enhanced. Once the pore radius is 50 Å, the region of anions is more compact and the density peak is enlarged, which implies that the external potential is dominated. Furthermore, the phenomena of spontaneous charge separation over -10 , -5 , and -2 V for $r = 20$, 35, and 50 Å are observed, respectively. Besides, it can be indicated that the ionic interfacial structure is closely related to the pore sizes, and the strength of ion correlations is reduced by the increase in the pore radius.

The variation of the charge density changes with distance for different pore radii is shown in Figure 6. The Coulomb force leads to the adsorption of counterions onto the electrode surface, resulting in an opposite charge peak near the wall. As the pore size becomes larger, the charge density near the wall continues to increase, and the charge-free layer occupied by carbon dioxide becomes wider.

Once the ion's charge density is determined, the potential distribution ϕ is calculated by integrating the discretized Poisson equation. In cylindrical systems, the Poisson equation can be rewritten as

$$\frac{1}{r} d \left(r \frac{d\phi}{dr} \right) = - \frac{\rho_e}{\epsilon_0} \quad (9)$$

where the charge density ρ_e can be calculated by the effect of all charge points in the carbon nanotube and the ϕ can be rewritten as follows

$$\phi(r) = - \frac{1}{\epsilon_0} \int_r^R s \rho_e(s) \ln \frac{s}{r} ds - \frac{\sigma}{\epsilon_0} R \ln \frac{R}{r} \quad (10)$$

the charge density ρ_e is discretized into 100 chunks along the radial direction; in each chunk, the ρ_e remains constant. Therefore, the first term of eq 10 is regarded as

$$\begin{aligned} \phi(r_1 - r_2) = & - \frac{\rho_e(r_1)}{\epsilon_0} \left(\frac{\ln r_2}{2} r_2^2 - \frac{\ln r_1}{2} r_1^2 - \frac{r_2^2 - r_1^2}{4} \right. \\ & \left. - \frac{r_2^2 - r_1^2}{2} \ln r_1 \right) \end{aligned} \quad (11)$$

By integrating from r_1 to R , the potential distribution $\phi(r_1)$ at a specific position r_1 can be simplified as

$$\phi(r_1) = \phi_{r_1-R} - \frac{\sigma}{\epsilon_0} R \ln \frac{R}{r_1} \quad (12)$$

where ϕ_{r_1-R} represents the value integrating from r_1 to R .

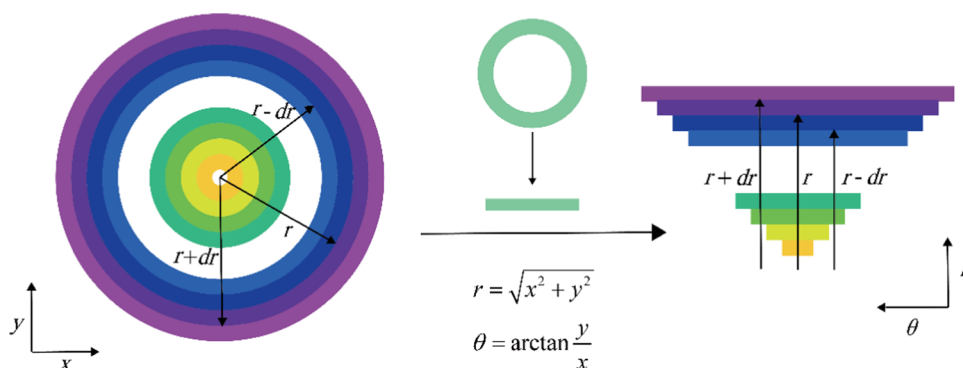


Figure 8. Illustrations of interacting molecule volumes in the Cartesian coordinate.

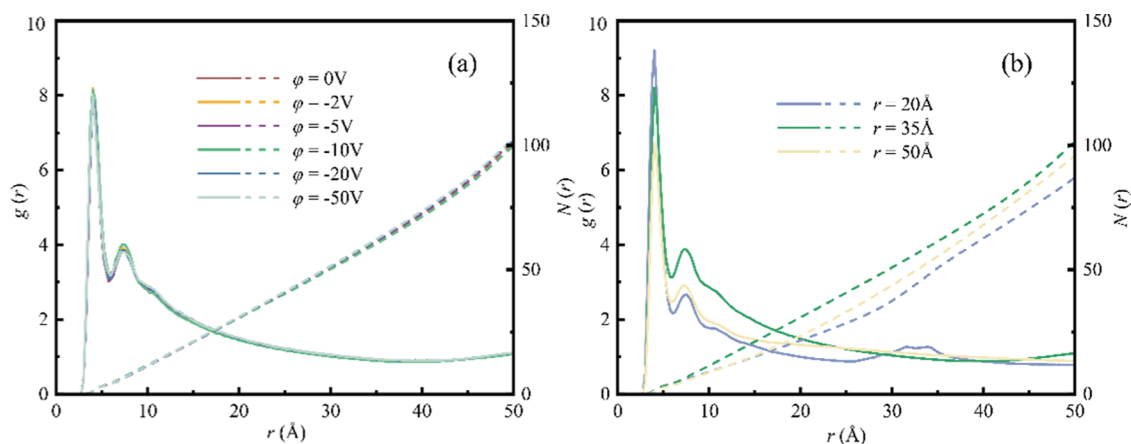


Figure 9. Radial distribution function and CN between carbon dioxide molecules at (a) $r = 35 \text{ Å}$ and (b) $\phi = -5 \text{ V}$.

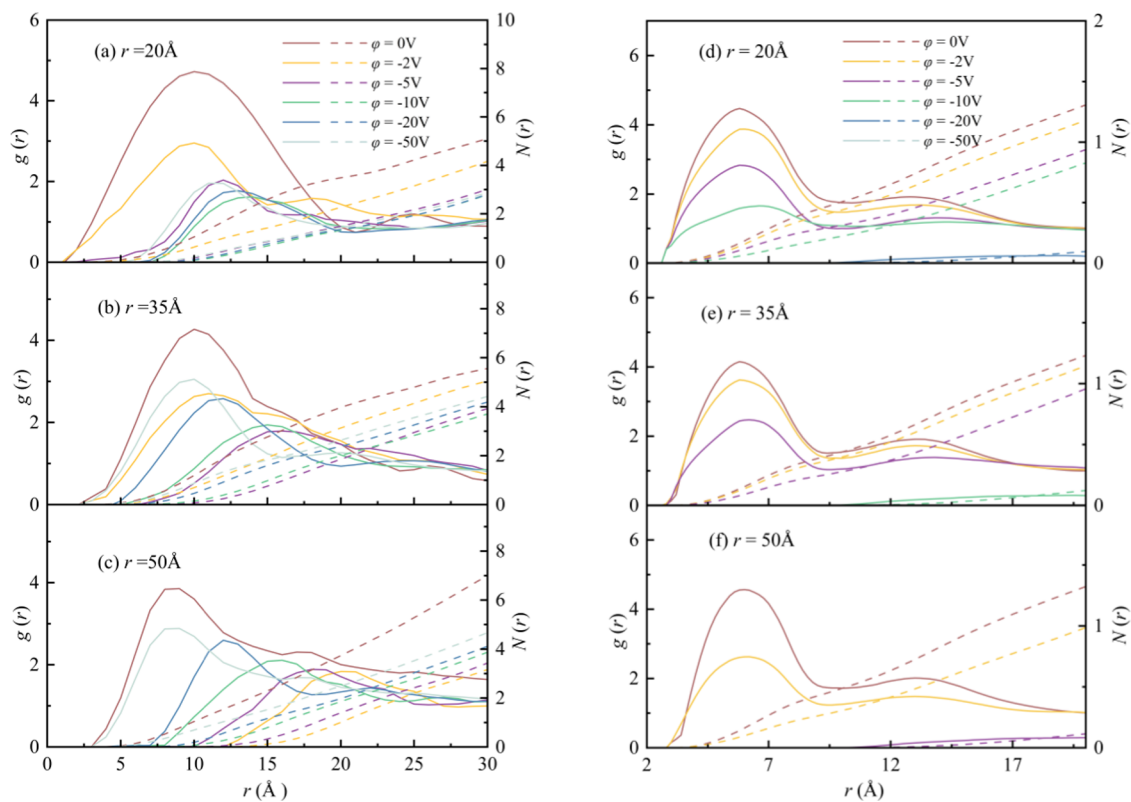


Figure 10. Radial distribution functions between cations and anions at pore sizes of (a) $r = 20 \text{ Å}$, (b) $r = 35 \text{ Å}$, and (c) $r = 50 \text{ Å}$. The radial distribution functions between cation and cation at pore sizes of (d) $r = 20 \text{ Å}$, (e) $r = 35 \text{ Å}$, and (f) $r = 50 \text{ Å}$, respectively.

The potential distribution φ varying along the radius direction together with the model schematic is shown in Figure 7a–c. At the electrode surface, the potential was initially set to 50 V and then rapidly decayed. When the distance from the electrode reaches 2.5 Å, the cations aggregate and begin to shield the external potential. As the distance exceeds 10 Å, the density profile in Figure 5 hardly changed, and the EDL dominated the potential distribution. So, the potential is constant and decreases to 0 V at the center of the carbon nanotube. Since the specific surface area of the nanotube increases quadratically with pore radius, the density of counterions attached to the surface of the tube wall increased rapidly, resulting in an enlarged positive local potential when $r = 50$ Å. The schematic diagram of surface counterions is depicted in Figure 7d. The internal charge distribution is irregular in the absence of an external potential. When external potential is applied, the cations with a positive charge move to the wall surface of the $r = 20$ Å to shield the negative charge at the electrode surface. Whereas, for the $r = 35$ Å, the counterions appear to be slightly overaggregated due to a larger specific surface area, and for $r = 50$ Å, the counterions are overaggregated.

3.2.1. Structure Evolution Process in the EDL. To further investigate the interfacial evolution and determine the spontaneous charge separation, the center of mass of each species is determined as the only point of determination as follows

$$i(x, y, z) = \left(\frac{\sum_{j \in i} x_j}{n}, \frac{\sum_{j \in i} y_j}{n}, \frac{\sum_{j \in i} z_j}{n} \right) \quad (13)$$

where x_j , y_j , and z_j represent the x , y , and z positions of each atom in the i th molecule, while n represents the relative atomic mass. On the other hand, the definition of RDF in eq 5 is defined to measure the probability density of the presence of other particles within a given distance, which is a thin region with radius r , as shown in Figure 8. To make the results more accurate, dr is set to 0.1 Å, and the relative position of each atom remains consistent.

By the recalculated methods, the RDF can be determined within the near-wall region, while the coordination numbers (CN) of particle solvation structures are calculated by integrating the RDF over the first peak. The RDF and CN between CO₂ under $r = 35$ Å are presented in Figure 9a, and the g hardly changes with the external potential. The location of the first peak was kept at $r = 4.3$ Å with the potential enhanced, owing to the electrical neutrality of CO₂. As for the effect of the pore radius, a similar situation can be observed; it indicates that the pore radius and the electrical interaction have played a negligible role in CO₂. The distance between CO₂ stays at 4.3 Å, while the CN equals about 95 all the time.

Once the variation of CO₂ can be ignored, the spatial structure caused by the cation in the near-wall region is described by partial distribution function (PDF). The PDF between [Bmim]⁺ and [Tf₂N][−] at different pore radii is depicted in Figure 10a–c. As the external field increases, the location of the peak remains constant and suddenly reduces, and the transition external potentials for $r = 20$ Å, $r = 35$ Å, and $r = 50$ Å are −10, −5, and −2 V, respectively. Consequently, the critical coordination numbers between cation and anion are 0.832, 0.951, and 0.994. It is consistent with the critical point determined by density distribution. The pair distributions of [Bmim]⁺ are compared between different external

potentials, as shown in Figure 10d–f. For a certain pore radius of 20 Å, at first, the location of the first peak gradually increases, while the value gradually decreases with the enhancement of the external potential, which implies that the repulsive force of the cation charge dominates. When the critically separated external potential is reached, the peak is at the farthest position, and the magnitude of the peak is at the lowest. At an external potential larger than the critical separation, the position of the first peak decreases slightly due to the external potential. Different pore sizes reported similar phenomena, except that the potential corresponding to the critical separation is different. On the other hand, the trend of the coordination number is similar to the above, and it also appears to become smaller first until the potential of the critical separation is followed by a larger one.

The coordination numbers of [Bmim]⁺ of the spatial structures are determined by integrating the pair distribution function over the first peak. Then, the CN of these species is used to estimate the coordination environment around each [Bmim]⁺ in the near-wall region. The ratio of coordination numbers at different external potentials and pore sizes is shown in Figure 11. Orange, green, and purple are uniquely

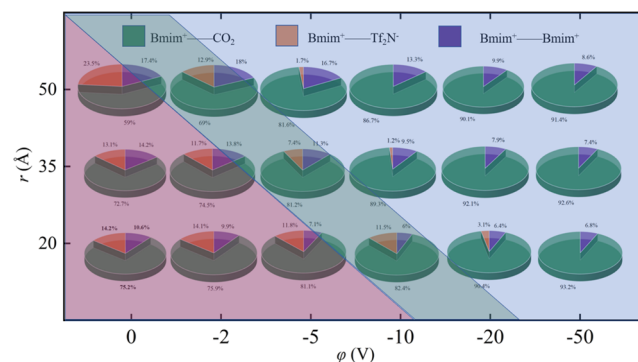


Figure 11. Ratio of the main cation coordination environment being affected by the external potential and pore radius.

contributed by the [Bmim]⁺ cation, CO₂, and [Tf₂N][−] anion, respectively. At a small pore size and external potential, the percent of cations corresponding to cations and anions is partially proportional. As the pore radius increases, the proportion of CO₂ corresponding to cations increases sharply due to the weakened confinement effect. Simultaneously, for the same pore sizes, the stronger external potential would enhance the distance between the anion and the electrode, so the proportion of anions decreases. The synergistic effect mechanism of the confinement effect and electrical force is evaluated by the critical separation points. Once the pore radius and external potential induced by the electrode exceed the critical points, the negative charge ion will be separated and move away from the electrode region.

The mobility of the electrode region of the species is evaluated through the variation of MSD. Figure 12a–c shows that MSD is linear with time. It can be found that the slope of the MSD at different pore sizes performs a first rise, and when it reaches the critical potential, there is a tendency for a sudden drop and then it continues to rise after that. This shows that before the critical potential, the agglomerated anions and cations start to separate, and the proportion of the cations moving toward the wall increases. After the critical potential, some cations are already immobile, while few cations stay

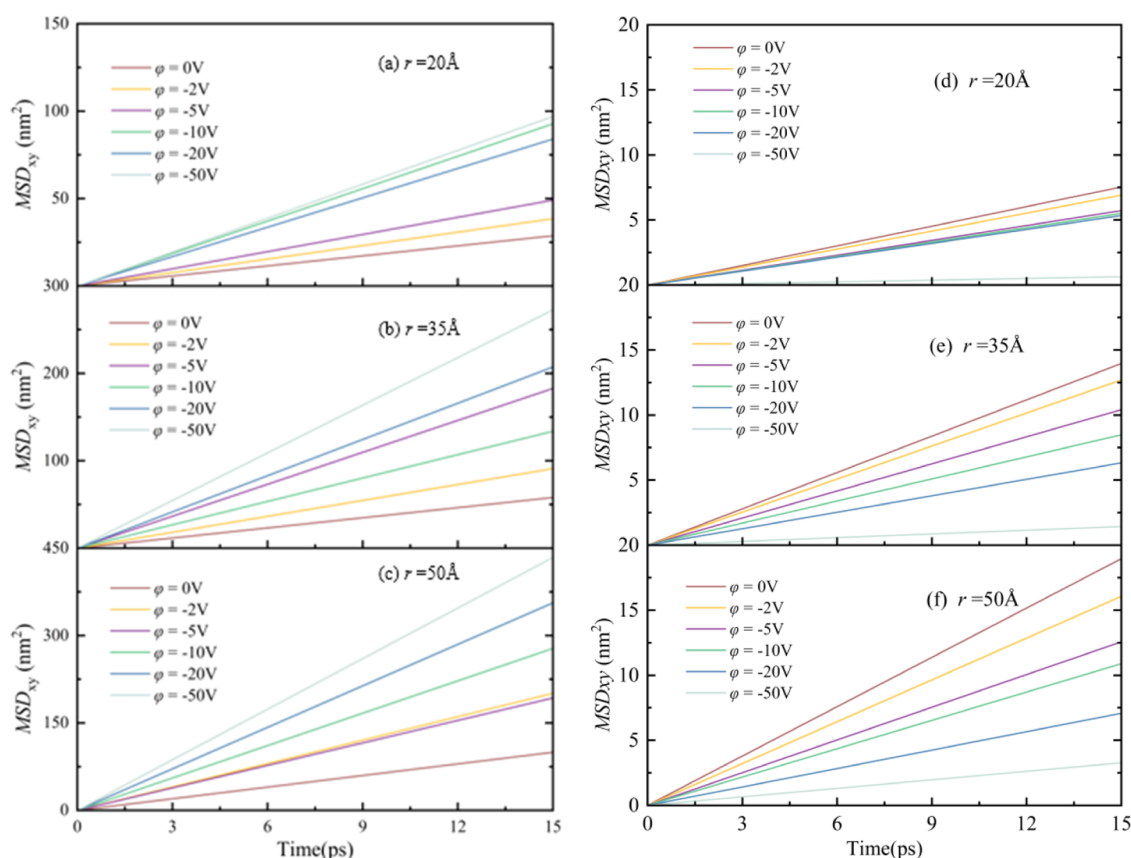


Figure 12. Mean square displacement of all cations at (a) $r = 20$ Å, (b) $r = 35$ Å, and (c) $r = 50$ Å and near-wall cations at (d) $r = 20$ Å, (e) $r = 35$ Å, (f) $r = 50$ Å.

active in the center of the tube. Subsequently, more and more cations detach from the anionic group with the increase of the potential. To better illustrate the effect of anions, Figure 12d–f shows the MSD of cations in the near-wall region versus time, which showed a linear variation trend, and the slope of the MSD decreases as the potential increases. At the larger external potential of -50 V, the slope of the MSD is much lower than that of -2 V, which suggests that the motion of the cations in the near-wall region is deduced from electrostatic interactions.

The variation of diffusion coefficients of cations at different potentials and pore sizes is shown in Figure 13, where the solid and dashed lines represent the diffusion coefficients of all cations and the D_{xy} of cations near the wall, respectively. By order of magnitude, the diffusion coefficients of ions near the wall are a few tenths of the diffusion coefficients of all cations, which indicates that, statistically, ions near the wall are rarely able to affect the diffusion coefficients of all ions. In other words, most of the diffusion coefficients of all ions are derived from the non-near-wall cations. It follows that the initial formation of the electric double layer has already begun after the transition external potential is reached, except that as the potential increases, more cations are separated from the anionic clusters. The evolution process of the EDL is closely related to the diffusion process of the cations. The diffusion coefficients of all cations increase with potential, and there is a local peak in the diffusion coefficients at the transition potential, which first decreases and then continues to rise. This indicates that more cations migrate as the potential increases, and the local peak indicates that the transition potential is the onset of anion separation.

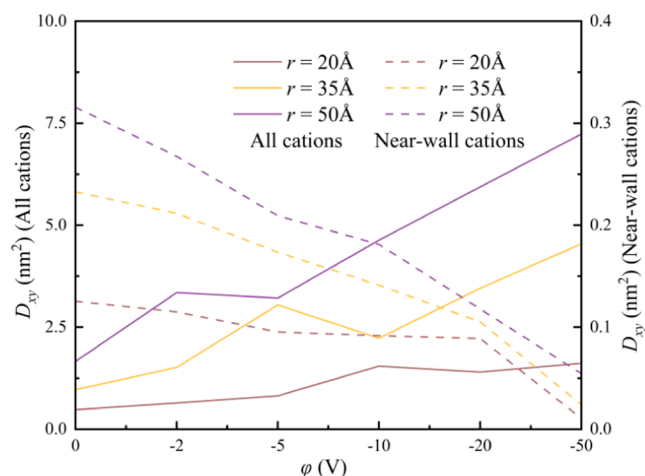


Figure 13. Variation of MSD and dynamics diffusion coefficients for [Bmim⁺] for different φ as a function of potential.

3.2.2. Polarization Evolution in EDL. As described before, when the external potential exceeds the critical potential at a certain radius, the EDL structure is dominated by the electrostatic force; otherwise, the confinement effect should be considered. To clarify the structural evolution in the near-wall region thoroughly, the orientation distribution of the [Bmim⁺] cation and the CO₂ molecule is analyzed.

Figure 14a–c presents the orientation distribution proportions of angle θ for CO₂ in the near-wall region, and the definition of θ is the angle between the O=C=O and the wall of the nanotube. Once the external potential is applied, the

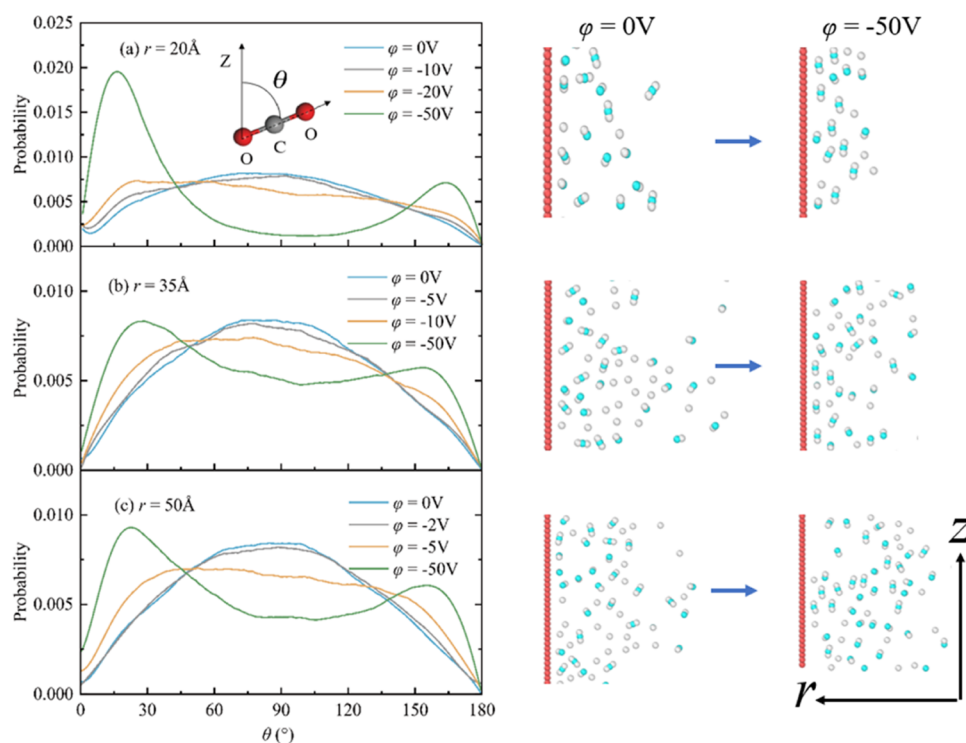


Figure 14. Schematic illustrations for the definitions of angle θ in the EDL of CO_2 at different external potentials for (a) $r = 20 \text{ \AA}$, (b) $r = 35 \text{ \AA}$, and (c) $r = 50 \text{ \AA}$, respectively.

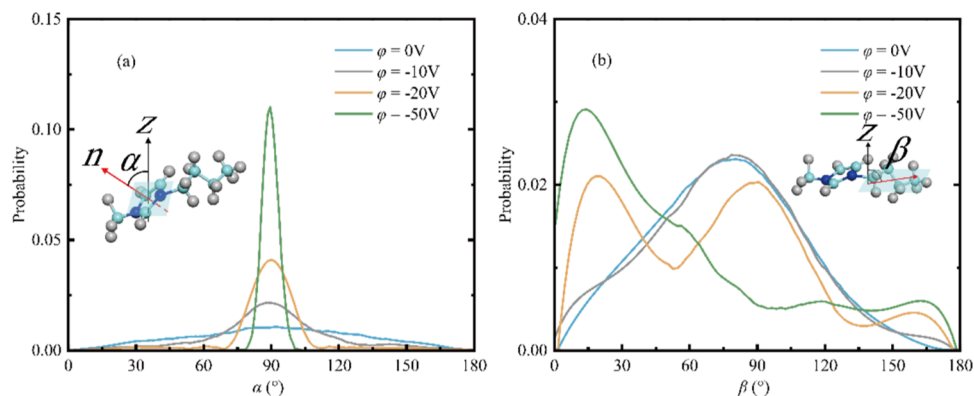


Figure 15. Schematic illustrations for the definitions of angles α and β of imidazole ring and alkyl chains planes in the EDL of $[\text{Bmim}^+]$ at different external potentials for $r = 20 \text{ \AA}$.

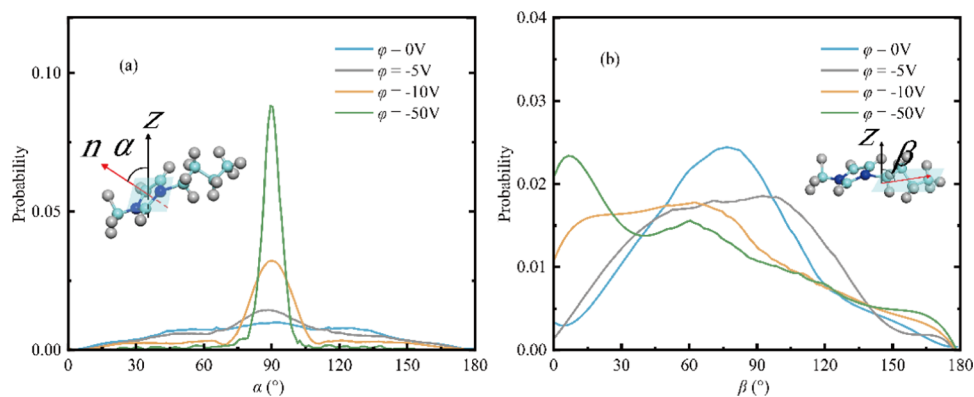


Figure 16. Schematic illustrations for the definitions of angles α and β of imidazole ring and alkyl chains planes in the EDL of $[\text{Bmim}^+]$ at different external potentials for $r = 35 \text{ \AA}$.

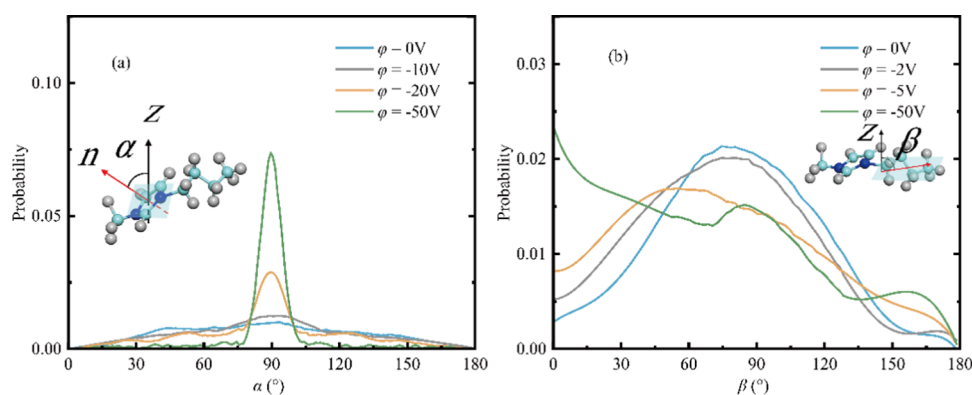


Figure 17. Schematic illustrations for the definitions of angles α and β of imidazole ring and alkyl chains planes in the EDL of $[\text{Bmim}^+]$ at different external potentials for $r = 50 \text{ \AA}$.

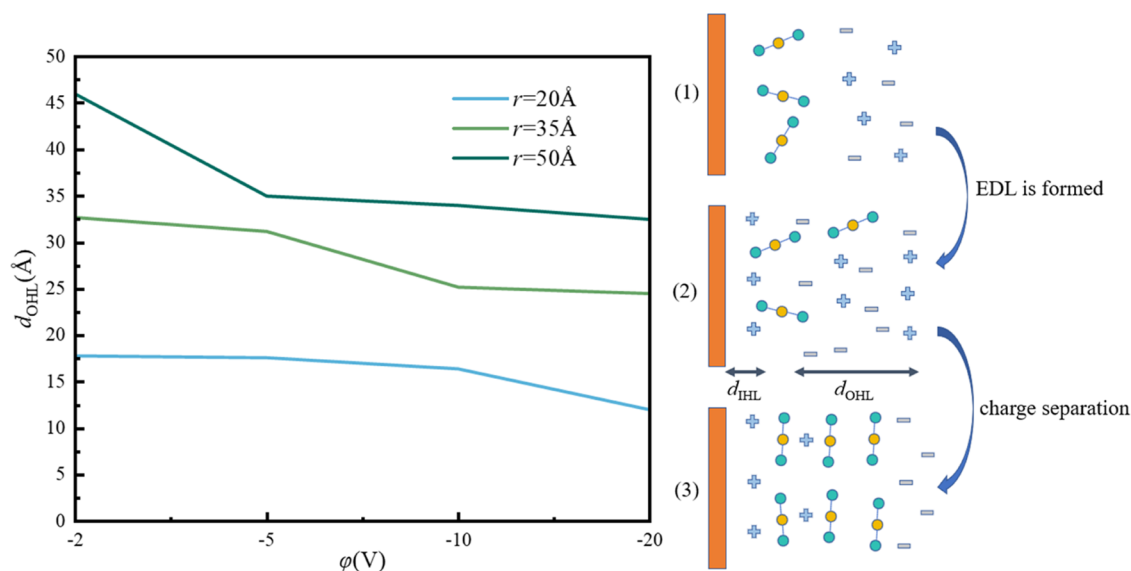


Figure 18. d_{OHL} with a change in the external potentials.

counterion $[\text{Bmim}^+]$ absorbed on the CNT will shield the attractive interaction of applying potential for CO_2 , so a disordering orientation of CO_2 can be observed. When the external potential becomes larger, the distribution of θ changes slowly. As the external potential exceeds the critical value, the counterions are insufficient to shield the influence of external potential; the parallel arrangement patterns can be observed, and two peaks at $\theta = 15$ and 165° are visible. A similar phenomenon has been found in previous research.⁴⁷ The MD snapshots of the CO_2 structure in the near-wall region at $r = 20 \text{ \AA}$ are presented on the right of Figure 14. When the external potential is less than the critical value, the CO_2 is arranged approximately vertically at the electrode interface; otherwise, the CO_2 molecules are preferentially parallel to the electrode surface.

To clarify the evolution orientation in the EDL, a vector of $[\text{Bmim}^+]$ is defined in this work, where n represents the normal vectors of imidazolium rings. Then, the orientational distribution of $[\text{Bmim}^+]$ at $r = 20 \text{ \AA}$ is evaluated by angles α and β in Figure 15, respectively. The $[\text{Bmim}^+]$ cations are randomly distributed at the electrode without an external potential, while the butyl surface is inclined in the vertical direction. As the external potential reaches the transition potential of -10 V , the imidazole ring is gradually oriented

preferentially perpendicular to the electrode surface. As for the terminal carbon atom of their alkyl chains, the single peak located at 90° is gradually changed to two peaks; finally, the peaks shift to $\theta = 20^\circ$ at a higher external potential, which implies the imidazole rings are oriented preferentially parallel to the electrode interface due to the external potential. The evolution process of $[\text{Bmim}^+]$ orientation distribution at $r = 35 \text{ \AA}$ and $r = 50 \text{ \AA}$ are presented in Figures 16 and 17, respectively, and the trend of change is consistent with $r = 20 \text{ \AA}$.

The outer Helmholtz double layer represents the outer plane where solvated ions from the bulk solution can approach the electrode; it can be determined by the location of anions and cations.⁴⁸ Figure 18 shows the variation of d_{OHL} with potential at different pore sizes. The length of the outer Helmholtz double layer d_{OHL} decreases with the increase of potential all the time. As the pore radius is $50 \text{ \AA}$, the inflection point appears at $\phi = -5 \text{ V}$, which is consistent with the division of the transition stage. Similar phenomena also can be observed at 35 and $20 \text{ \AA}$. In addition, the schematic of the variation of the EDL structure has been presented on the right of Figure 18. When a stable EDL is formed, the thickness of the EDL no longer changes dramatically and the potential at this time begins to regulate the direction of ions and molecules.

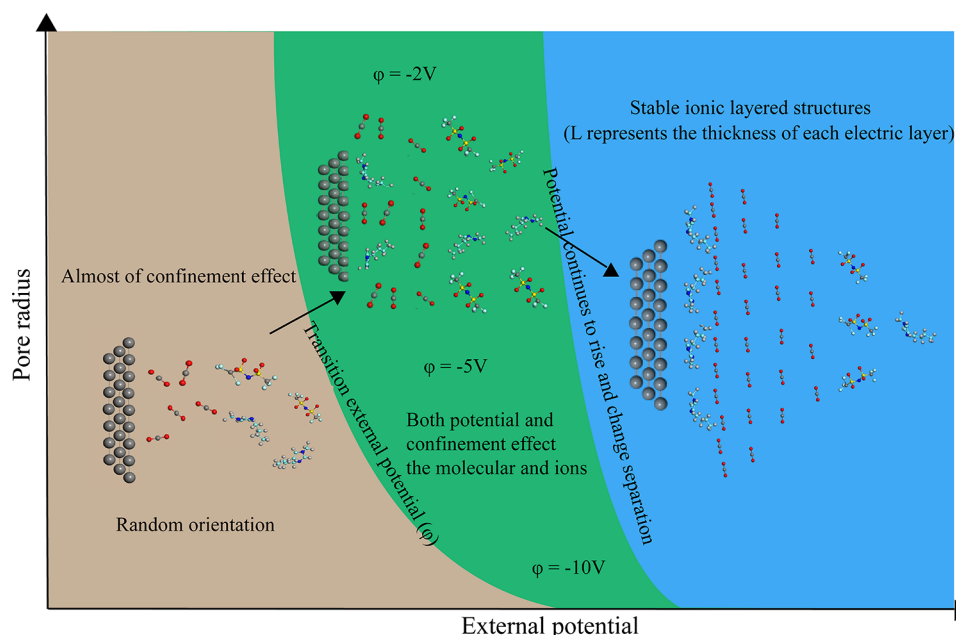


Figure 19. Schematic presentation of the formation and transition mechanisms of the electric double layer and molecular orientation.

The EDL formation depends on the polarization and the electrolyte. The external potential dominates the polarization, while the electrolyte contains IL and CO₂, both of which are affected by the confinement. Herein, the process of forming the electric double-layer structure and the evolution of the molecular orientation by the variation of the external potential and the confinement effect caused by the pore radius are synergistic, as displayed in Figure 19. The structural changes in EDL are divided into three stages. In the first stage, CO₂ aggregates as small molecules on the walls due to the dominance of confinement effects; the density of CO₂ varies very little with the potential due to the dominance of confinement effects. In the second stage, a state in which some cations coaggregate with CO₂ occurs due to the Coulomb force and the van der Waals force; in addition, cations also need to overcome the attraction of anions. In the last stage, there is an occupation of the near-wall surface by pure cations, and the phenomenon of complete separation of charges begins to appear. With the increases of pore sizes, a relatively lower potentials are needed for the stage changes.

4. CONCLUSIONS

Using MD simulation, the structure and orientation properties of IL/CO₂ in the vicinity of a carbon electrode with several external potentials were studied by the constant potential method. Herein, three different pore radii of CNTs are considered: 20, 35, and 50 Å, respectively. The following significant findings are presented.

- (1) Due to the confinement effect, both the CO₂ and ion liquids are aggregated in the vicinity of the electrode. With the increase of pore radius, the nonelectrostatic force is weakened. Thus, the cation [Bmim⁺] is closer to the wall, while the CO₂ near the surface is pushed out of the EDL. When the critical potential is reached, charge separation phenomena are observed. The occurrence of transition points for certain pores decays with the decrease of pore radius.

- (2) During the transition stage, the coordination micro-environment of [Bmim⁺] in the vicinity of the electrode is traced constantly. As the pore radius increases, the coordination number proportion of [Bmim⁺] and [Tf₂N⁻] is enlarged. As the external potential enhances, the coordination number is occupied by [Bmim⁺] in the parallel plane and CO₂ in the vertical direction, and the increased ordering of species can be observed. Once the transition process is completed, the steady micro-environment is presented in EDL.
- (3) The mobility of IL/CO₂ and the orientation distribution are consistent with the transition process. When the transition process happens, the diffusion coefficient of [Bmim⁺] decreases slightly with the variation of external potential. Meanwhile, the orientation of CO₂ changes from vertical to parallel to the electrode, while the imidazole ring of [Bmim⁺] is gradually oriented preferentially perpendicular to the electrode surface. From a molecular view, the structure characteristic validates the EDL transition process.

In summary, the structural evolution of the EDL involves the local concentration of CO₂, the interfacial electric field, and the ion distribution. By focusing on the variation of the internal structural aspects of the EDL with potential and pore size, this paper may provide the molecular understanding of the collective effect of thermodynamics and kinetics properties on CO₂.

■ ASSOCIATED CONTENT

Data Availability Statement

Data will be made available on request.

■ AUTHOR INFORMATION

Corresponding Author

Xiang Wei – Hebei Key Laboratory of Low Carbon and High Efficiency Power Generation Technology, North China Electric Power University, Baoding, Hebei 071003, China;

orcid.org/0009-0003-4890-4412; Email: xiangwei@ncepu.edu.cn

Authors

Hongwei Chen – Hebei Key Laboratory of Low Carbon and High Efficiency Power Generation Technology, North China Electric Power University, Baoding, Hebei 071003, China

Yiming Han – Hebei Key Laboratory of Low Carbon and High Efficiency Power Generation Technology, North China Electric Power University, Baoding, Hebei 071003, China

Xiaoxiao Hu – Hebei Key Laboratory of Low Carbon and High Efficiency Power Generation Technology, North China Electric Power University, Baoding, Hebei 071003, China

Yangfan Song – Hebei Key Laboratory of Low Carbon and High Efficiency Power Generation Technology, North China Electric Power University, Baoding, Hebei 071003, China

Zhuo Liu – Hebei Key Laboratory of Low Carbon and High Efficiency Power Generation Technology, North China Electric Power University, Baoding, Hebei 071003, China

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acs.jpcc.4c00619>

Author Contributions

The manuscript was written with contributions from all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (No. 52306083) and the Fundamental Research Funds for the Central Universities (No. 2023MS119).

ABBREVIATIONS

CO₂, carbon dioxide; IL, ionic liquid; [Bmim⁺][Tf₂N[−]], 1-Butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide; RDF, radial distribution function; PDF, pair distribution function; DFT, density functional theory; CO₂RR, reduction reaction of carbon dioxide; FCM, fixed charge method; CPM, constant potential method; PNP, Poisson–Nernst–Planck; GCS, Gouy–Chapman–Stern; EDL, electric double layer

REFERENCES

- (1) Hwang, J. Y.; Myung, S. T.; Myung, S. T.; Sun, Y. K. Sodium-ion batteries: present and future. *Chem. Soc. Rev.* **2017**, *46* (12), 3529–3614.
- (2) Miller, J. R.; Outlaw, R.; Holloway, B. Graphene double-layer capacitor with ac line-filtering performance. *Science* **2010**, *329* (5999), 1637–1639.
- (3) Koohi-Fayegh, S.; Rosen, M. A. A review of energy storage types, applications and recent developments. *J. Energy Storage* **2020**, *27*, No. 101047.
- (4) Fan, X.; Zhang, W.; Xu, Y.; Zheng, J.; Li, Y.; Fan, X.; Zhang, F.; Ji, J.; Peng, W. Porous structure engineering of N-doped carbons for enhanced mass transfer towards High-Performance supercapacitors and Li-Ion batteries. *J. Colloid Interface Sci.* **2022**, *624*, 51–59.
- (5) Jiang, H.; Luo, R.; Li, Y.; Chen, W. Recent advances in solid–liquid–gas three-phase interfaces in electrocatalysis for energy conversion and storage. *EcoMater* **2022**, *4* (4), No. 12199, DOI: 10.1002/eom2.12199.

- (6) Pan, F.; Yang, Y. Designing CO₂ reduction electrode materials by morphology and interface engineering. *Energy Environ. Sci.* **2020**, *13* (8), 2275–2309.
- (7) Ren, Y.; Yu, C.; Tan, X.; et al. Strategies to suppress hydrogen evolution for highly selective electrocatalytic nitrogen reduction: challenges and perspectives. *Energy Environ. Sci.* **2021**, *14* (3), 1176–1193.
- (8) Zhou, S. How ion size influences energy storage in cylindrical nanoporous supercapacitors. *J. Phys. Chem. C* **2019**, *123* (49), 29638–29646.
- (9) Xu, G.; Han, J.; Ding, B.; Nie, P.; Pan, J.; Dou, H.; Li, H.; Zhang, X. Biomass-derived porous carbon materials with sulfur and nitrogen dual-doping for energy storage. *Green Chem.* **2015**, *17* (3), 1668–1674.
- (10) Deng, X.; Li, J.; Ma, L.; Sha, J.; Zhao, N. Three-dimensional porous carbon materials and their composites as electrodes for electrochemical energy storage systems. *Mater. Chem. Front.* **2019**, *3* (11), 2221–2245.
- (11) Liang, L.; Qi, Y.; Xue, F.; Bhattacharya, S.; Harris, S. J.; Chen, L. Q. Nonlinear phase-field model for electrode-electrolyte interface evolution. *Phys. Rev. E* **2012**, *86* (5), No. 051609.
- (12) Ravikovitch, P. I.; Haller, G. L.; Neimark, A. V. Density functional theory model for calculating pore size distributions: pore structure of nanoporous catalysts. *Adv. Colloid Interface Sci.* **1998**, *76–77*, 203–226.
- (13) Wang, X.; Liu, K.; Wu, J. Demystifying the Stern layer at a metal–electrolyte interface: Local dielectric constant, specific ion adsorption, and partial charge transfer. *J. Chem. Phys.* **2021**, *154*, No. 124701.
- (14) Doblhoff-Dier, K.; Koper, M. T. Modeling the Gouy–Chapman diffuse capacitance with attractive ion–surface interaction. *J. Phys. Chem. C* **2021**, *125* (30), 16664–16673.
- (15) Oldham, K. B. A Gouy–Chapman–Stern model of the double layer at a (metal)/(ionic liquid) interface. *J. Electroanal. Chem.* **2008**, *613* (2), 131–138.
- (16) Gongadze, E.; Van Rienen, U.; Iglíč, A. Generalized stern models of the electric double layer considering the spatial variation of permittivity and finite size of ions in saturation regime. *Cell. Mol. Biol. Lett.* **2011**, *16*, 576–594.
- (17) Wang, H.; Pilon, L. Physical interpretation of cyclic voltammetry for measuring electric double layer capacitances. *Electrochim. Acta* **2012**, *64*, 130–139.
- (18) Islam, M. M.; Alam, M. T.; Ohsaka, T. Electrical double-layer structure in ionic liquids: a corroboration of the theoretical model by experimental results. *J. Phys. Chem. C* **2008**, *112* (42), 16568–16574.
- (19) Tenney, C. M.; Massel, M.; Mayes, J. M.; Sen, M.; Brennecke, J. F.; Maginn, E. J. A computational and experimental study of the heat transfer properties of nine different ionic liquids. *J. Chem. Eng. Data* **2014**, *59* (2), 391–399.
- (20) Fayon, P.; Thomas, J.; Trewin, A. Structure and properties of a nanoporous supercapacitor. *J. Phys. Chem. C* **2016**, *120* (45), 25880–25891.
- (21) Li, Z.; Jeanmairat, G.; Méndez-Morales, T.; Rotenberg, B.; Salanne, M. Capacitive performance of water-in-salt electrolytes in supercapacitors: A simulation study. *J. Phys. Chem. C* **2018**, *122* (42), 23917–23924.
- (22) Jiang, D.-e.; Wu, J. Microscopic insights into the electrochemical behavior of nonaqueous electrolytes in electric double-layer capacitors. *J. Phys. Chem. Lett.* **2013**, *4* (8), 1260–1267.
- (23) Wang, Q.; Xie, H.; Hu, Z.; Liu, C. The impact of the electric field on surface condensation of water vapor: Insight from molecular dynamics simulation. *Nanomaterials* **2019**, *9* (1), No. 64, DOI: 10.3390/nano9010064.
- (24) Yamada, S. A.; Hung, S. T.; Thompson, W. H.; Fayer, M. D. Effects of pore size on water dynamics in mesoporous silica. *J. Chem. Phys.* **2020**, *152*, No. 154704.
- (25) Winarto; Yamamoto, E.; Yasuoka, K. Water molecules in a carbon nanotube under an applied electric field at various temper-

- atures and pressures. *Water* **2017**, *9* (7), No. 473, DOI: 10.3390/w9070473.
- (26) Noh, C.; Jung, Y. Understanding the charging dynamics of an ionic liquid electric double layer capacitor via molecular dynamics simulations. *Phys. Chem. Chem. Phys.* **2019**, *21* (13), 6790–6800.
- (27) Yang, H.; Yang, J.; Bo, Z.; Zhang, S.; Yan, J.; Cen, K. Edge effects in vertically-oriented graphene based electric double-layer capacitors. *J. Power Sources* **2016**, *324*, 309–316.
- (28) Wang, Z.; Yang, Y.; Olmsted, D. L.; Asta, M.; Laird, B. B. Evaluation of the constant potential method in simulating electric double-layer capacitors. *J. Chem. Phys.* **2014**, *141*, No. 184102.
- (29) Yang, L.; Fishbine, B. H.; Migliori, A.; Pratt, L. R. Molecular simulation of electric double-layer capacitors based on carbon nanotube forests. *J. Am. Chem. Soc.* **2009**, *131* (34), 12373–12376.
- (30) Reed, S. K.; Lanning, O. J.; Madden, P. A. Electrochemical interface between an ionic liquid and a model metallic electrode. *J. Chem. Phys.* **2007**, *126*, No. 084704.
- (31) Merlet, C.; Rotenberg, B.; Madden, P. A.; Taberna, P.-L.; Simon, P.; Gogotsi, Y.; Salanne, M. On the molecular origin of supercapacitance in nanoporous carbon electrodes. *Nat. Mater.* **2012**, *11* (4), 306–310.
- (32) Merlet, C.; Salanne, M.; Rotenberg, B.; Madden, P. A. Imidazolium ionic liquid interfaces with vapor and graphite: interfacial tension and capacitance from coarse-grained molecular simulations. *J. Phys. Chem. C* **2011**, *115* (33), 16613–16618.
- (33) Su, J.; Musgrave, C. B., III; Song, Y.; et al. Strain enhances the activity of molecular electrocatalysts via carbon nanotube supports. *Nat. Catal.* **2023**, *6* (9), 818–828.
- (34) Heinz, H.; Vaia, R.; Farmer, B.; Naik, R. R. Accurate simulation of surfaces and interfaces of face-centered cubic metals using 12–6 and 9–6 Lennard-Jones potentials. *J. Phys. Chem. C* **2008**, *112* (44), 17281–17290.
- (35) Uddin, M.; Coombe, D. Kinetics of CH₄ and CO₂ hydrate dissociation and gas bubble evolution via MD simulation. *J. Phys. Chem. A* **2014**, *118* (11), 1971–1988.
- (36) Pan, G.-C.; Ding, J.; Wang, W.; et al. Molecular simulations of the thermal and transport properties of alkali chloride salts for high-temperature thermal energy storage. *Int. J. Heat Mass Transfer* **2016**, *103*, 417–427.
- (37) Fumi, F.; Tosi, M. Ionic sizes and born repulsive parameters in the NaCl-type alkali halides—I: The Huggins-Mayer and Pauling forms. *J. Phys. Chem. Solids* **1964**, *25* (1), 31–43.
- (38) Wennberg, C. L.; Murtola, T.; Páll, S.; et al. Direct-space corrections enable fast and accurate Lorentz–Berthelot combination rule Lennard-Jones lattice summation. *J. Chem. Theory Comput.* **2015**, *11* (12), 5737–5746.
- (39) Shim, Y.; Kim, H. J. MD study of solvation in the mixture of a room-temperature ionic liquid and CO₂. *J. Phys. Chem. B* **2010**, *114* (31), 10160–10170.
- (40) Aghaie, M.; Rezaei, N.; Zendejboudi, S. New insights into bulk and interface properties of [Bmim][Ac]/[Bmim][BF₄] ionic liquid/CO₂ systems—Molecular dynamics simulation approach. *J. Mol. Liq.* **2020**, *317*, No. 113497.
- (41) Bi, S.; Banda, H.; Chen, M.; et al. Molecular understanding of charge storage and charging dynamics in supercapacitors with MOF electrodes and ionic liquid electrolytes. *Nat. Mater.* **2020**, *19* (5), 552–558.
- (42) Feng, G.; Qiao, R.; Huang, J.; et al. The importance of ion size and electrode curvature on electrical double layers in ionic liquids. *Phys. Chem. Chem. Phys.* **2011**, *13* (3), 1152–1161.
- (43) Pardoux, É.; Veretennikov, Y. On the Poisson equation and diffusion approximation. I. *Ann. Probab.* **2001**, *29* (3), 1061–1085, DOI: 10.1214/aop/1015345596.
- (44) Wu, Q.; McDowell, M. T.; Qi, Y. Effect of the Electric Double Layer (EDL) in Multicomponent Electrolyte Reduction and Solid Electrolyte Interphase (SEI) Formation in Lithium Batteries. *J. Am. Chem. Soc.* **2023**, *145* (4), 2473–2484.
- (45) Wang, Z.; Chen, J.; Li, Y.; Dong, K.; Yu, Y. EDL structure of ionic liquid-MXene-based supercapacitor and hydrogen bond role on

the interface: a molecular dynamics simulation investigation. *Phys. Chem. Chem. Phys.* **2022**, *24* (10), 5903–5913.

(46) Ravikumar, B.; Mynam, M.; Rai, B. Effect of salt concentration on properties of lithium ion battery electrolytes: a molecular dynamics study. *J. Phys. Chem. C* **2018**, *122* (15), 8173–8181.

(47) Qin, J.; Wang, M.; Wang, Y.; et al. Understanding Electric Field-Dependent Structure Variation of Functional Ionic Liquids at the Electrode Interface. *ChemElectroChem* **2021**, *8* (9), 1588–1595.

(48) Kong, X.; Lu, D.; Liu, Z.; Wu, J. Molecular dynamics for the charging behavior of nanostructured electric double layer capacitors containing room temperature ionic liquids. *Nano Res.* **2015**, *8*, 931–940.



CAS BIOFINDER DISCOVERY PLATFORM™

**CAS BIOFINDER
HELPS YOU FIND
YOUR NEXT
BREAKTHROUGH
FASTER**

Navigate pathways, targets, and
diseases with precision

Explore CAS BioFinder

